CoachApp: A Workout Tracker with a Built-in AI Agent

Project #2 Proposal – Agentic System

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Abstract

CoachApp is a web app for logging workouts. Its AI agent reads the user's training history, shows where progress is being made, and suggests how to adjust future training.

1 Problem

Most workout trackers only store numbers. Users must read the charts themselves and decide alone what to change. Beginners in particular do not know if they are progressing or when to increase the weight, keep it the same, or rest.

2 Proposed Solution

CoachApp has two parts:

- **Tracker:** the user logs workouts (exercise, sets, reps, weight) and sees their history and simple charts.
- **AI agent:** the user asks, for example, "How am I doing?" and the agent analyses the data and answers with a short summary and a suggestion.

3 The AI Agent

3.1 What it does

1. **Shows progress.** For example: "Your bench press went up 10 kg in 6 weeks" or "Your squat has not improved for 3 weeks."
2. **Suggests adjustments.** For example: "Add 2.5 kg next session" or "Take a lighter week."
3. **Keeps the user in control.** Suggestions are proposals; the user accepts or rejects them.

3.2 Why an agent and not just charts?

Charts only display data. To answer "How am I doing?" the system must look up the right history, calculate trends, compare weeks, and decide what to recommend. This takes several steps and uses tools, which is what makes it an agent rather than a simple chatbot.

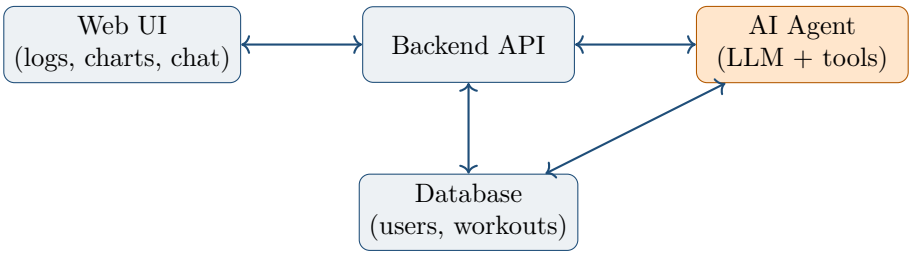
3.3 Tools

Tool	Purpose
`get_history`	Read past workouts (by date range or exercise)
`get_progress`	Calculate trends: weekly volume, personal records, plateaus
`suggest_adjustment`	Create a proposed change to the training (needs user approval)

3.4 Example flow

User asks "How am I doing?" → the agent calls `get_history` → calls `get_progress` → writes a summary → calls `suggest_adjustment` → the user approves or rejects.

4 Architecture



Proposed stack: React frontend, Python (FastAPI) backend, SQLite/PostgreSQL database, and an LLM API with tool calling (e.g. Claude API).

5 Scope

MVP (must have):

- User login
- Workout logging with a form
- History page and progress chart
- Agent chat: progress summary and adjustment suggestions

Possible later additions: logging workouts by typing a sentence, weekly plan generation, weekly summary messages, CSV export.